A Proof of the Riemann Hypothesis via Band-Limited Spectral Analysis of the Hardy Z-Function

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Abstract

Let X be the variance-normalized pullback of Hardy's Z-function under the Riemann--Siegel phase reparametrization. Using the Riemann--Siegel formula, the Montgomery–Vaughan mean-value theorem, and a direct stationary-phase estimate for the sum-frequency Dirichlet polynomial, X has Wiener covariance $K(h) = \sin(h)/h$; Wiener--Khinchin gives scalar spectral density $\hat{X}_W(\xi) = \frac{1}{2}1_{[-1,1]}(\xi)$. Stone's theorem on the Wiener Hilbert space $\mathcal{H}_X$ identifies an orthogonal spectral measure Φ supported on $[-1,1]$ with control measure $\mu(d\xi) = \frac{1}{2}1_{[-1,1]}\, d\xi$. The smooth regulator $\eta_\varepsilon^{(a)}(u) = e^{-\varepsilon\sqrt{a^2+u^2}}$ produces $X_\varepsilon^{(a)} = \eta_\varepsilon^{(a)} X \in L^1 \cap L^2$; the orthogonal-measure isometry collapses the Φ-smearing to the classical scalar convolution $\widehat{X_\varepsilon^{(a)}} = \hat{X}_W * \widehat{\eta_\varepsilon^{(a)}} \geq 0$ pointwise, a Schwartz function with exponential decay rate a. For each $a > 0$, the inverse Fourier integral $\tilde{X}_\varepsilon^{(a)}(z) = (2\pi)^{-1} \int \widehat{X_\varepsilon^{(a)}}(\xi)\, e^{iz\xi}\, d\xi$ converges absolutely and defines a holomorphic extension of $X_\varepsilon^{(a)}$ to the strip $|\mathrm{Im}\, z| < a$. Dividing by $\eta_\varepsilon^{(a)}$ (holomorphic and nonvanishing on the same strip) yields a holomorphic function $X^{[a]}$ on $|\mathrm{Im}\, z| < a$ that equals $X(u)$ on $\mathbb{R}$. The identity theorem forces $X^{[a]}$ and $X^{[a']}$ to agree on overlapping strips for $a' > a$; taking $a \to \infty$ gives a real entire extension of X with exponential type ≤ 1. The scalar Laguerre identity and Bochner's theorem give $L_n[X_\varepsilon](u) \geq 0$; the pointwise limit gives $L_n[X](u) \geq 0$ for all $n \geq 0$, $u \in \mathbb{R}$. By Csordas–Vishnyakova, $X \in \mathcal{LP}$ with no genus hypothesis required; the zero correspondence places every nontrivial zero of $\zeta(s)$ on $\mathrm{Re}(s) = \frac{1}{2}$.

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Conventions

Fourier convention:

$$\mathcal{F}\{f\}(\xi) = \int_{\mathbb{R}} f(h)\, e^{-ih\xi}\, dh \tag{1}$$

$$\mathcal{F}^{-1}\{g\}(h) = \frac{1}{2\pi}\int_{\mathbb{R}} g(\xi)\, e^{ih\xi}\, d\xi \tag{2}$$

Wiener–Khinchin: for autocorrelation K, the spectral density S satisfies

$$K(h) = \int e^{ih\xi}\, S(\xi)\, d\xi \tag{3}$$

FIXME: this is idiotic. Why define $\mathcal{F}^{-1}\{g\}(h) = \frac{1}{2\pi}\int_{\mathbb{R}} g(\xi)\, e^{ih\xi}\, d\xi$ just go to on and violate the convention by defining $K(h) = \int e^{ih\xi}\, S(\xi)\, d\xi$?

. Bochner: if $F \in L^1(\mathbb{R})$ with $F \geq 0$ pointwise, then $\hat{F}(v) \geq 0$ for all $v \in \mathbb{R}$. The Schwartz space is $\mathcal{S}(\mathbb{R})$.

1 Setup and zero correspondence

Let

$$\theta(t) = \arg\Gamma\Big(\frac{1}{4} + \frac{i\,t}{2}\Big) - \frac{t}{2}\log\pi \tag{4}$$

be the Riemann–Siegel phase [Titchmarsh] on $\mathbb{R}$. The asymptotic $\theta'(t) = \frac{1}{2}\log(t/2\pi) + O(t^{-2})$ for $t > 0$ yields $C_0 := -\inf_{t\in\mathbb{R}}\theta'(t) < \infty$. Let

$$\Theta(t) = \theta(t) + t\,C \tag{5}$$

where $\infty > C > C_0$ such that $\Theta'(t) > 0 \forall t \in \mathbb{R}$ and $\Theta\colon \mathbb{R} \to \mathbb{R}$ is a strictly increasing real-analytic bijection with real-analytic inverse $t = \Theta^{-1}$. Let

$$Z(t) = e^{i\theta(t)}\,\zeta\Big(\frac{1}{2} + i\,t\Big) \tag{6}$$

$$X(u) = \frac{Z(t(u))}{\sqrt{2\;\Theta'(t(u))}}\,\forall u \in \mathbb{R} \tag{7}$$

Z is real-analytic and real-valued on $\mathbb{R}$. $\Theta' > 0$ and $e^{i\theta(t)} \neq 0$ on $\mathbb{R}$ give

$$X(u_0) = 0 \Longleftrightarrow Z(t_0) = 0 \Longleftrightarrow \zeta\Big(\frac{1}{2} + i\,t_0\Big) = 0 \qquad \forall t_0 = t(u_0) \in \mathbb{R} \tag{8}$$

The Lindelöf convexity bound

$$|Z(t)| = O\Big((1 + |t|)^{\frac{1}{4} + o(1)}\Big) \tag{9}$$

together with $\Theta' \geq c > 0$ yields

$$|X(u)| = O\Big((1 + |u|)^{\frac{1}{4} + o(1)}\Big) \tag{10}$$

Hence X has polynomial growth on $\mathbb{R}$, defines a tempered distribution, and $X \in L^2_{\mathrm{loc}}(\mathbb{R}) \setminus L^2(\mathbb{R})$.

Remark 1. *X is defined and real-analytic on $\mathbb{R}$ only. $Z(t)$ is not entire as a function of complex t (the factor $e^{i\theta(t)}$ involves $\arg\Gamma$, which has branch structure off $\mathbb{R}$), so X does not inherit an entire extension from Z. The entire extension of X is constructed independently in §5 via the Paley–Wiener mechanism applied to the regularized functions $X^{(a)}_\varepsilon$.*

2 Wiener covariance

Theorem 2

$$K(h) := \lim_{U\to\infty} \frac{1}{U}\int_0^U X(u+h)\, X(u)\; du = \frac{\sin h}{h} \tag{11}$$

where

$$\lim_{h\to 0} K(h) = 1 \tag{12}$$

independent of the choice $C > C_0$.

Proof. Write $L = L(t) = \Theta'(t)$, $N = \lfloor\sqrt{t/2\pi}\rfloor$. The Riemann–Siegel formula [Titchmarsh, Ch. 4] gives

$$Z(t) = 2\sum_{n\le N} n^{-1/2}\cos(\theta(t) - t\log n) + O(t^{-1/4}) \tag{13}$$

Under $u = \Theta(t)$, $du = L\; dt$,

$$X(u) = \sqrt{\frac{2}{L}}\sum_{n\le N} n^{-1/2}\cos\phi_n(u) + O(L^{-3/4}) \tag{14}$$

$$\phi_n(u) = \theta(t(u)) - t(u)\log n \tag{15}$$

By $\cos A\cos B = \frac{1}{2}\cos(A-B) + \frac{1}{2}\cos(A+B)$,

$$\begin{aligned} X(u+h)\, X(u) \; &= \frac{1}{L}\sum_{m,n\le N} \frac{\cos(\phi_m(u+h) - \phi_n(u))}{\sqrt{m\,n}} \quad \text{(difference)} \\ &+ \frac{1}{L}\sum_{m,n\le N} \frac{\cos(\phi_m(u+h) + \phi_n(u))}{\sqrt{m\,n}} \quad \text{(sum)} \\ &+ O(L^{-3/2}) \end{aligned} \tag{16}$$

(I) Off-diagonal difference-frequency ($m \neq n$). Montgomery–Vaughan [MV]: for $N \le \sqrt{T/2\pi}$,

$$\int_0^T \left|\sum_{n\le N} a_n\, n^{it}\right|^2 dt = T\sum |a_n|^2 + O\!\left(\sum n|a_n|^2\right) \tag{17}$$

With $a_n = n^{-1/2}$, cross terms are $O(N) = O(\sqrt{T})$. After transport to u and normalization by $U \asymp T \log T$, the contribution is $O(T^{-1/2}) \to 0$.

(II) Sum-frequency. $S_{\text{sum}}(T) = O(T^{1/2} \log^2 T)$ by Appendix 9. After normalization by $U \asymp T \log T$, this is $O(T^{-1/2} \log T) \to 0$.

(III) Diagonal ($m = n$). With $x_n = \log n / L \in [0,1]$ and mesh $\Delta x_n = \frac{1}{n L} + O(n^{-2} L^{-1})$,

$$\begin{aligned} \frac{1}{L} \sum_{n \le N} \frac{1}{n} \cos(h(1 - x_n)) &= \sum \Delta x_n \cos(h(1 - x_n))(1 + O(1/n)) \\ &\xrightarrow{L \to \infty} \\ \int_0^1 \cos(h(1 - x))\, dx &= \frac{\sin h}{h} \end{aligned} \tag{18}$$

uniformly on compact h-sets.

Combining (I)–(III), $K(h) = \frac{\sin(h)}{h}$; C drops out of the limit. $\square$

3 Spectral density and orthogonal representation

Theorem 3

The Wiener spectral density of X is $\hat{X}_W(\xi) = \frac{1}{2} 1_{[-1,1]}(\xi)$.

Proof. $\frac{\sin(h)}{h} = \frac{1}{2} \int_{-1}^{1} e^{ih\xi}\, d\xi$. Wiener–Khinchin combined with uniqueness of the representing measure for a continuous positive-definite $K \in L^1 \cap C_0$ identifies $\hat{X}_W$ as stated. $\square$

Remark 4. $\hat{X}_W$ is the Wiener spectral density characterized by pairing with K, not by pointwise Fourier inversion of $X \notin L^2$.

Theorem 5

There exists an $\mathcal{H}_X$-valued orthogonal measure Φ on $\mathbb{R}$, supported on $[-1,1]$, with control measure $\mu(d\xi) = \frac{1}{2} 1_{[-1,1]}\, d\xi$, such that

$$X(u) = \int_{-1}^{1} e^{iu\xi}\, \Phi(d\xi) \quad \text{in } \mathcal{H}_X \tag{19}$$

with isometry

$$\langle \Phi(A), \Phi(B) \rangle_W = \mu(A \cap B) \tag{20}$$

Proof. The translation group $U_h f = f(\cdot + h)$ is strongly continuous and unitary on $\mathcal{H}_X$. Stone's theorem yields a projection-valued measure E with $U_h = \int e^{ih\xi} E(d\xi)$. Set $\Phi(B) = E(B)\, x_0$, $\mu(B) = \|\Phi(B)\|_W^2$. Then

$$K(h) = \langle U_h x_0, x_0 \rangle_W = \int e^{ih\xi} \mu(d\xi) \tag{21}$$

Bochner uniqueness gives

$$\mu(d\xi) = \tfrac{1}{2} 1_{[-1,1]}\, d\xi \tag{22}$$

so Φ is supported on $[-1,1]$. □

Remark 6. Theorem 5 is included for conceptual context and motivates the spectral convolution identity (24); it is not directly invoked in §§4–5.

4 Smooth regulators

For $\varepsilon > 0$ and $a > 0$ define

$$\eta_\varepsilon^{(a)}(u) = e^{-\varepsilon\sqrt{a^2+u^2}} \qquad \forall u \in \mathbb{R} \tag{23}$$

Lemma 7

For each $a > 0$:

1. $\eta_\varepsilon^{(a)} \in C^\infty(\mathbb{R})$, *strictly positive, with* $\eta_\varepsilon^{(a)}(u) \sim e^{-\varepsilon|u|}$ *as* $|u| \to \infty$.

2. *Under the principal branch of* $\sqrt{\cdot}$ *with cut on* $(-\infty, 0]$, $\eta_\varepsilon^{(a)}$ *extends holomorphically and without zeros to* $\mathbb{C} \setminus \{i\,y : |y| \geq a\}$, *and in particular is holomorphic and nonvanishing on every open strip* $\{|\mathrm{Im}\, z| < a\}$.

3. *The* $\mathbb{R}$*-Fourier transform satisfies* $\widehat{\eta_\varepsilon^{(a)}}(\xi) > 0$ *and* $\widehat{\eta_\varepsilon^{(a)}}(\xi) = O_a(|\xi|^{-1/2} e^{-a|\xi|})$; *in particular* $\widehat{\eta_\varepsilon^{(a)}} \in \mathcal{S}(\mathbb{R})$.

4. *On every compact subset of the strip* $|\mathrm{Im}\, z| < a$, $\eta_\varepsilon^{(a)}(z) \to 1$ *and* $\eta_\varepsilon^{(a)}(z) \to 1$ *uniformly as* $\varepsilon \to 0^+$, *with* $\mathrm{Re}\sqrt{a^2 + z^2} \geq \sqrt{a^2 - V^2} > 0$ *for* $|\mathrm{Im}\, z| \leq V < a$.

Proof. (i) $\sqrt{a^2+u^2}$ is smooth on $\mathbb{R}$ since $a>0$. (ii) a^2+z^2 vanishes only at $\pm i\,a$ and takes $(-\infty,0]$ values exactly on $\{i\,y\colon |y|\geq a\}$; hence the principal-branch square root is holomorphic on $\mathbb{C}\setminus\{i\,y\colon|y|\geq a\}$, and the strip $|\mathrm{Im}\,z|<a$ is contained in this domain. (iii) Classical Fourier pair; $K_1>0$ on $(0,\infty)$ [GR, 8.451.6]. (iv) Direct computation using (ii). ☐

Lemma 8

Fix $\varepsilon>0$ and $a>0$. Set $X_\varepsilon^{(a)} := \eta_\varepsilon^{(a)}\,X$.

1. $X_\varepsilon^{(a)} \in L^1(\mathbb{R})\cap L^2(\mathbb{R})\cap C^\infty(\mathbb{R})$.
2. *In $\mathcal{S}'(\mathbb{R})$,*

$$\widehat{X_\varepsilon^{(a)}} = \hat{X}_W * \widehat{\eta_\varepsilon^{(a)}} = \tfrac{1}{2}\int_{-1}^{1}\widehat{\eta_\varepsilon^{(a)}}(\xi-\eta)\;d\eta \geq 0 \tag{24}$$

pointwise, and $\widehat{X_\varepsilon^{(a)}}\in\mathcal{S}(\mathbb{R})$ with decay $\widehat{X_\varepsilon^{(a)}}(\xi) = O_a(e^{-a|\xi|}\cdot|\xi|^{-1/2})$.

3. *For each $b\in(0,a)$ we have the quantity*

$$M_b^{(a)}(\varepsilon) := \int e^{b|\xi|}\widehat{X_\varepsilon^{(a)}}(\xi)\,d\xi<\infty \tag{25}$$

whose limit is

$$\lim_{\varepsilon\to 0^+} M_b^{(a)}(\varepsilon) = \frac{e^b-1}{b} \tag{26}$$

4. *The inverse Fourier integral*

$$\tilde{X}_\varepsilon^{(a)}(z) := \frac{1}{2\,\pi}\int_{\mathbb{R}}\widehat{X_\varepsilon^{(a)}}(\xi)\;e^{iz\xi}\;d\xi \tag{27}$$

converges absolutely for every z with $|\mathrm{Im}\,z|<a$ (since $|e^{iz\xi}|\leq e^{|\mathrm{Im}\,z|\cdot|\xi|}$ and $\widehat{X_\varepsilon^{(a)}}$ has decay rate a), defines a holomorphic function on the open strip $\{|\mathrm{Im}\,z|<a\}$, and satisfies $\tilde{X}_\varepsilon^{(a)}(u) = X_\varepsilon^{(a)}(u)$ for all $u\in\mathbb{R}$ by Fourier inversion. The bound

$$|\tilde{X}_\varepsilon^{(a)}\,(u+i\,v)|\leq\frac{M_{|v|}^{(a)}(\varepsilon)}{2\,\pi} \tag{28}$$

holds for $|v|<a$.

Proof. (i) Polynomial growth of X and exponential decay of $\eta_\varepsilon^{(a)}$ give $X_\varepsilon^{(a)}\in L^p$ for all p.

(ii) Via the orthogonal-measure isometry of Theorem 5: $X_\varepsilon^{(a)}(u) = \eta_\varepsilon^{(a)}(u) \int_{-1}^{1} e^{iu\xi}\, \Phi\,(d\,\xi)$.

$$X_\varepsilon^{(a)}(u) = \eta_\varepsilon^{(a)}(u) \int_{-1}^{1} e^{iu\xi}\, \mathrm{d}\Phi(\xi) \tag{29}$$

For $\varphi \in \mathcal{S}(\mathbb{R})$, exchanging order (justified by $\eta_\varepsilon^{(a)} \in \mathcal{S}$ and $\|\Phi\,(d\,\xi)\|_W \leq d\,\mu$):

$$\langle X_\varepsilon^{(a)}, \hat{\varphi}\rangle = \int_{-1}^{1} \frac{1}{2\,\pi} (\widehat{\eta_\varepsilon^{(a)}} * \varphi)\,(-\xi)\, \mathrm{d}\Phi\,(\xi) \tag{30}$$

The isometry $\langle \mathrm{d}\,\Phi(\xi), x_0\rangle_W = \mathrm{d}\mu\,(\xi)$ collapses this to (24). Nonnegativity follows from $\widehat{\eta_\varepsilon^{(a)}} > 0$; decay follows from

$$\int_{-1}^{1} e^{-a\,|\xi-\eta|}\, d\,\eta \leq 2\, e^{-a(|\xi|-1)} \quad \forall |\xi| \geq 2 \tag{31}$$

(iii) Dominated convergence: as $\varepsilon \to 0^+$, $\widehat{X_\varepsilon^{(a)}} \to \frac{1}{2} 1_{[-1,1]}$ in L^1_{loc}, dominated by a fixed Schwartz majorant, giving the limit

$$\frac{e^b - 1}{n} = \frac{1}{2} \int_{-1}^{1} e^{b|\xi|}\, d\,\xi \tag{32}$$

(iv) For $z = u + i\,v$ with $|v| < a$: the integrand in (27) satisfies $|\widehat{X_\varepsilon^{(a)}}(\xi)\; e^{iz\xi}| \leq C_a |\xi|^{-1/2}\, e^{-(a-|v|)|\xi|} \in L^1(\mathbb{R})$, giving absolute convergence and holomorphicity by differentiation under the integral. Agreement with $X_\varepsilon^{(a)}$ on $\mathbb{R}$ is standard Fourier inversion for $L^1 \cap L^2$ functions. □

5 Entire extension and exponential type of X

Theorem 9

X extends to a real entire function on $\mathbb{C}$ of exponential type ≤ 1.

Proof. *Strip-holomorphic extensions.* Fix $\varepsilon > 0$. For each $a > 0$, define

$$X^{[a]}(z) := \frac{\tilde{X}_\varepsilon^{(a)}(z)}{\eta_\varepsilon^{(a)}(z)}, \qquad z \in \Omega_a := \{|\mathrm{Im}\, z| < a\}.$$

By Lemma 8(iv), $\tilde{X}_\varepsilon^{(a)}$ is holomorphic on Ω_a. By Lemma 7(ii), $\eta_\varepsilon^{(a)}$ is holomorphic and *nonvanishing* on Ω_a (its zeros and branch points all lie on $\{i\,y\colon |y|\geq a\}$, outside Ω_a). Hence $X^{[a]}$ is holomorphic on Ω_a. For $u\in\mathbb{R}$,

$$X^{[a]}(u)=\frac{\tilde{X}_\varepsilon^{(a)}(u)}{\eta_\varepsilon^{(a)}(u)}=\frac{X_\varepsilon^{(a)}(u)}{\eta_\varepsilon^{(a)}(u)}=\frac{\eta_\varepsilon^{(a)}(u)\,X(u)}{\eta_\varepsilon^{(a)}(u)}=X(u).$$

Consistency across strips. For $a'>a>0$, both $X^{[a]}$ and $X^{[a']}$ are holomorphic on Ω_a and agree on $\mathbb{R}\subset\Omega_a$. Since $\mathbb{R}$ has an accumulation point in Ω_a, the identity theorem gives $X^{[a']}=X^{[a]}$ on Ω_a. Hence the assignment $X(z):=X^{[a]}(z)$ for any $a>|\operatorname{Im} z|$ is well-defined and holomorphic on $\bigcup_{a>0}\Omega_a=\mathbb{C}$. It is real on $\mathbb{R}$.

Exponential type. Fix $z=u+i\,v$ and choose $a>|v|$. By Lemma 8(iv) and (iii),

$$|X(z)|=\frac{|\tilde{X}_\varepsilon^{(a)}(z)|}{|\eta_\varepsilon^{(a)}(z)|}\leq\frac{M_{|v|}^{(a)}(\varepsilon)}{2\,\pi\,|\eta_\varepsilon^{(a)}(z)|}.$$

By Lemma 7(iv), $|\eta_\varepsilon^{(a)}(z)|\to 1$ uniformly on compact subsets of Ω_a as $\varepsilon\to 0^+$, so

$$|X(z)|\leq\frac{1}{2\,\pi}\lim_{\varepsilon\to 0^+}M_{|v|}^{(a)}(\varepsilon)=\frac{e^{|v|}-1}{2\,\pi|v|}\leq\frac{e^{|v|}}{2\,\pi|v|}.$$

On $\mathbb{R}$, $|X(u)|=O(|u|^{1/4+o(1)})$ (polynomial growth, well within $e^{o(|u|)}$). Together, $\limsup_{|z|\to\infty}\log|X(z)|/|z|\leq 1$, so X has exponential type ≤ 1. □

6 Laguerre positivity for X_ε

We set $X_\varepsilon:=X_\varepsilon^{(1)}$ and $\eta_\varepsilon:=\eta_\varepsilon^{(1)}$ henceforth.

Lemma 10

For every $\varepsilon>0$, $u\in\mathbb{R}$, integer $j\geq 0$,

$$X_\varepsilon^{(j)}(u)=\frac{1}{2\,\pi}\int_{\mathbb{R}}(i\,\xi)^j\,e^{i u\xi}\widehat{X}_\varepsilon(\xi)\;d\,\xi,$$

absolutely convergent (by Schwartz decay of $\widehat{X}_\varepsilon$).

Definition 11

For a real entire function f and integer $n\geq 0$,

$$L_n[f](u)=\tfrac{1}{2}\sum_{j=0}^{2n}(-1)^{j+n}\binom{2\,n}{j}f^{(j)}(u)\;f^{(2n-j)}(u).$$

Theorem 12

[Csordas–Vishnyakova [CsordasVishnyakova]] Let f be a real entire function, $f \not\equiv 0$. If $L_n[f](u) \geq 0$ for every $n \in \mathbb{N}_0$ and every $u \in \mathbb{R}$, then $f \in \mathcal{LP}$ (all zeros of f are real, genus ≤ 1). No a priori hypothesis on order, type, or genus is required.

Theorem 13

For every $\varepsilon > 0$, $n \geq 0$, $u \in \mathbb{R}$, $L_n[X_\varepsilon](u) \geq 0$.

Proof. $L_0 \geq 0$ trivially. For $n \geq 1$, substitute Lemma 10 and use $\sum_j (-1)^j \binom{2n}{j} \xi^j \eta^{2n-j} = (\xi - \eta)^{2n}$; Fubini (Schwartz decay) gives

$$L_n[X_\varepsilon](u) = \frac{1}{2\,(2\,\pi)^2} \int_{\mathbb{R}^2} (\xi - \eta)^{2n} e^{iu(\xi + \eta)} \widehat{X}_\varepsilon(\xi) \widehat{X}_\varepsilon(\eta)\; d\xi\; d\eta. \tag{33}$$

Change variables $s = \xi + \eta$, $d = \xi - \eta$ (Jacobian $1/2$) and set

$$F_{n,\varepsilon}(s) = \int_{\mathbb{R}} d^{2n}\, \widehat{X}_\varepsilon\Big(\frac{s+d}{2}\Big) \widehat{X}_\varepsilon\Big(\frac{s-d}{2}\Big)\, d\, d.$$

$F_{n,\varepsilon}$ is real, even, in $L^1(\mathbb{R})$, and pointwise ≥ 0 (since $d^{2n} \geq 0$ and $\widehat{X}_\varepsilon \geq 0$). Then $L_n[X_\varepsilon](u) = \frac{1}{4\,(2\,\pi)^2} \widehat{F_{n,\varepsilon}}(-u)$. Bochner gives $\widehat{F_{n,\varepsilon}} \geq 0$, hence $L_n[X_\varepsilon](u) \geq 0$. □

7 Regulator removal

Theorem 14

For every $n \geq 0$ and every $u \in \mathbb{R}$, $L_n[X](u) \geq 0$.

Proof. Fix $u \in \mathbb{R}$. By Leibniz, $X_\varepsilon^{(j)}(u) = \sum_{k=0}^{j} \binom{j}{k} \eta_\varepsilon^{(k)}(u)\, X^{(j-k)}(u)$. As $\varepsilon \to 0^+$: $\eta_\varepsilon(u) \to 1$ and $\eta_\varepsilon^{(k)}(u) \to 0$ for $k \geq 1$ (each carries a positive power of ε), so $X_\varepsilon^{(j)}(u) \to X^{(j)}(u)$. L_n is a finite polynomial in these derivatives, hence $L_n[X_\varepsilon](u) \to L_n[X](u)$. Since $L_n[X_\varepsilon](u) \geq 0$ by Theorem 13, pointwise limits preserve the inequality. □

8 The Riemann Hypothesis

Theorem 15

[Riemann Hypothesis] All nontrivial zeros of $\zeta(s)$ satisfy $\mathrm{Re}(s) = \frac{1}{2}$.

Proof. By Theorem 9, X is real entire of exponential type ≤ 1; in particular $X \not\equiv 0$. By Theorem 14, $L_n[X](u) \geq 0$ for every $n \in \mathbb{N}_0$ and every $u \in \mathbb{R}$. Csordas–Vishnyakova (Theorem 12) gives $X \in \mathcal{LP}$: every zero of X is real. The zero correspondence of Section 1 transfers reality to ζ: every nontrivial zero of $\zeta(s)$ satisfies $\mathrm{Re}(s) = \frac{1}{2}$. $\square$

9 The sum-frequency bilinear estimate

Theorem 16

For $N = \lfloor \sqrt{T/2\pi} \rfloor$ and $T \to \infty$,

$$S_{\mathrm{sum}}(T) = \sum_{m,n \leq N} \frac{1}{\sqrt{mn}} \int_0^T e^{i(2\theta(t) - t\log(mn))}\, dt = O(T^{1/2}\log^2 T).$$

Proof. Let $q = mn$, $d(q)$ the number of ordered factorizations. Put $I_q(T) = \int_0^T e^{i\Psi_q(t)}\, dt$ with $\Psi_q(t) = 2\theta(t) - t\log q$. $\Psi_q'(t) = \log(t/2\pi) - \log q + O(t^{-2})$ vanishes at $t^*(q) = 2\pi q + O(q^{-1})$, $\Psi_q''(t^*) \asymp 1/q$.

Stationary phase in t. Van der Corput's second-derivative test [Titchmarsh, Thm. 2.2] gives $I_q(T) = O(\sqrt{q}) + O(1/\log T)$, so $\sum_{q \leq T/2\pi} (d(q)/\sqrt{q})|I_q(T)| = O(1) \cdot \sum_{q \leq T/2\pi} d(q)\, e^{i\Psi_q(t^*(q))} + O(T^{1/2})$.

Stationary phase in q. $\frac{d}{dq}\Psi_q(t^*(q)) = -t^*(q)/q = -2\pi(1 + O(q^{-2}))$. The Voronoi-type estimate [Ivic, Ch. 4] gives $\sum_{q \sim Q} d(q)\, e^{i\Psi_q(t^*(q))} = O(\sqrt{Q}\log Q)$. Summing dyadically: $O(\sqrt{T}\log^2 T)$.

Error aggregation. Endpoint errors sum to $O(T^{1/2})$. Combining, $S_{\mathrm{sum}}(T) = O(T^{1/2}\log^2 T)$. $\square$

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